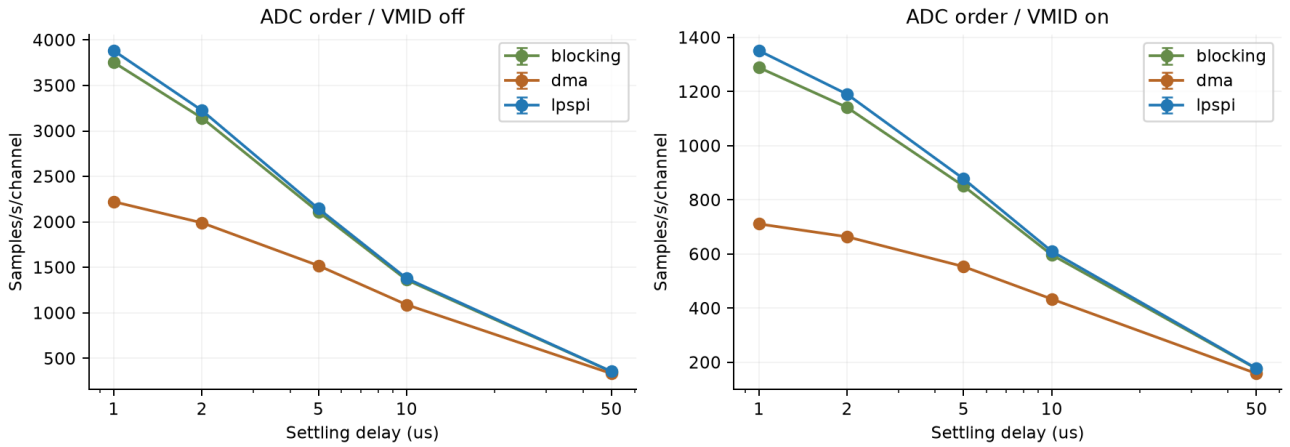
TestBoard_ADC124
Timing budget and SPI engine comparison

13 October 2026 | Both full arrays | 16 MHz requested SPI | Manual / repeat 1

Production sampling rate: three engines

Same open-connector, charger-connected setup and bias resistors as the settling report. ADC-order scan, five settling delays and VMID off/on. Three fresh repetitions per setting: 1 s warm-up + 5 s measurement. Rates use device timestamps, not host arrival or disk-write time. Diagnostic firmware results are separate.

Settle us	VMID	Blocking Hz/ch	DMA Hz/ch	LPSPI Hz/ch	LPSPI vs blocking	LPSPI vs DMA
1	off	3,753.7	2,222.2	3,882.5	3.4%	74.7%
1	on	1,288.9	711.2	1,351.3	4.8%	90.0%
2	off	3,142.9	1,990.6	3,225.8	2.6%	62.1%
2	on	1,141.6	663.8	1,190.5	4.3%	79.3%
5	off	2,109.7	1,519.8	2,145.9	1.7%	41.2%
5	on	851.0	553.6	877.6	3.1%	58.5%
10	off	1,363.4	1,090.2	1,379.2	1.2%	26.5%
10	on	597.0	433.7	609.9	2.2%	40.6%
50	off	355.4	333.6	356.5	0.3%	6.9%
50	on	176.2	158.6	177.3	0.6%	11.8%



Cells are medians of three repetitions; graph error bars show the repetition range. One sweep = one value for each of 50 sensor routes. Aggregate useful values/s = 50 x Hz/channel. All three engines use the same shared SPI bus and acquire the two ADCs sequentially.

What happens in one sweep

Order	VMID	Groups	Actual address changes	Prime words	Useful words	VMID words	Total words
ADC	off	50	52	50	50	0	100
ADC	on	50	100	50	50	250	350
Address	off	14	16	14	50	0	64
Address	on	14	28	14	50	70	134

ADC order: array, MUX / ADC input, external input. Each of the 50 sensor routes forms its own address group. Each group clocks one selector prime and one retained sample. VMID-on additionally parks the MUX quad and clocks five discarded words (one prime + four ADC inputs).

Address order (firmware scanorder mux): seven external addresses per array; addresses 0-5 read four ADC inputs, and address 6 reads only MUX4. One shared selector prime per group reduces the sensor+prime word count from 100 to 64. All 50 physical routes are still measured.

VMID-off adds two real park changes per sweep, one at the array handoff and one at the end. VMID-on already parks after each group, so those two additional park calls return without changing GPIOs or waiting. Counting calls rather than actual address changes would overestimate VMID-on settling by 2 x delay.

Operation	Nominal component at 16 MHz	Evidence
MUX change	EN off; 200 ns; 3 address writes; EN on; 200 ns; settling delay	Firmware sequence
GPIO writes	5 writes per actual address change	Firmware sequence
ADC track window	3 SCLK cycles = 0.1875 us	TI datasheet
Conversion / serial output	13 SCLK cycles = 0.8125 us	TI datasheet
Complete ADC word	16 SCLK cycles = 1.0000 us	Clock-derived
CS allowances	100 ns setup + 100 ns high per word	Firmware delays
USB frame	4-byte header + 100-byte values + 10-byte timing = 114 bytes	Protocol

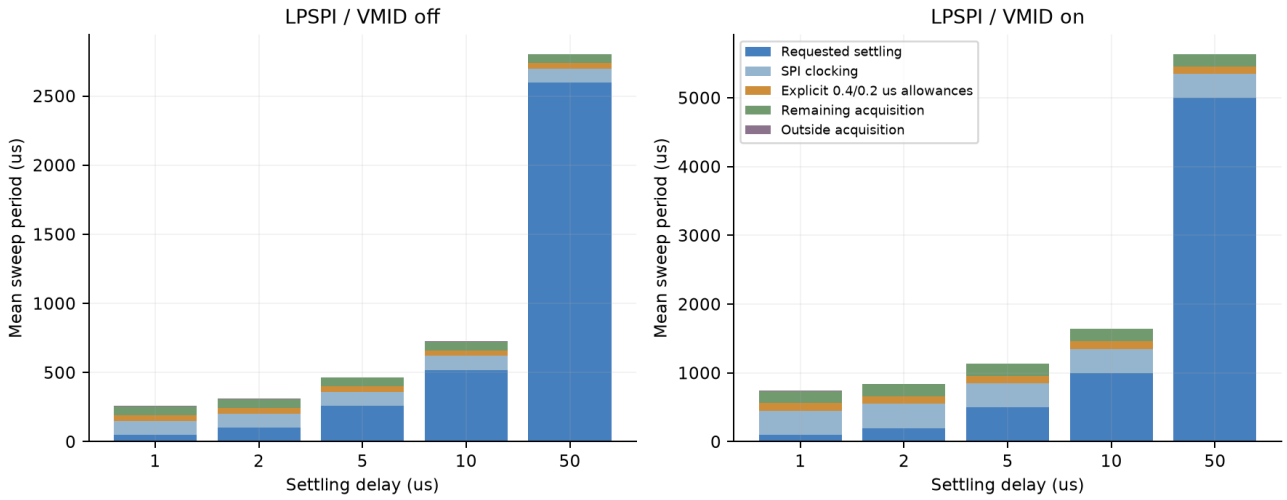
ADC tracking, conversion and data shifting occupy the same 16-clock frame; they must not be added again as separate delays. The DIN selector applies to the following conversion. These ADC timing facts follow TI ADC124S101 datasheet SNAS283D, sections Operation / Using the ADC and timing specifications. TI datasheet.

The pinned SPI library configures a 240 MHz peripheral root and divider 15 for a 16 MHz request. Timing calculations use nominal 16 MHz; SCLK and GPIO edge times were not independently scoped. The 0.4 us MUX allowance is a firmware wait budget, not a measured electrical switch propagation time.

Calculated production timing budget: LPSPI

T = 1,000,000 / rate = acquisition + outside-acquisition gap. Acquisition contains requested settling, SPI clocking, explicit MUX/CS allowances, and a remaining cost. The remaining cost includes GPIO/register work, batch/transfer loops, delay-call overhead and interrupts; it is not a uniquely identified phase.

Settle us	VMID	Settling us	SPI clock us	Explicit waits us	Remaining acquisition us	Gap us	Total period us
1	off	52.00	100.00	40.80	63.76	1.00	257.56
1	on	100.00	350.00	110.00	178.85	1.17	740.02
2	off	104.00	100.00	40.80	63.83	1.37	310.00
2	on	200.00	350.00	110.00	179.00	1.00	840.00
5	off	260.00	100.00	40.80	64.20	1.00	466.00
5	on	500.00	350.00	110.00	178.37	1.12	1139.50
10	off	520.00	100.00	40.80	63.14	1.10	725.04
10	on	1000.00	350.00	110.00	178.36	1.12	1639.48
50	off	2600.00	100.00	40.80	63.23	1.12	2805.15
50	on	5000.00	350.00	110.00	178.31	1.13	5639.43



Settling = 52 x delay with VMID off, or 100 x delay with it on. SPI clocking = 100 or 350 us. Explicit allowance totals = 52 x 0.4 + 100 x 0.2 = 40.8 us off, or 100 x 0.4 + 350 x 0.2 = 110.0 us on. These are additive nominal budgets; the residual closes the measured accounting.

Rows use median repetition means. All CSV calculations retain full precision. Acquisition is measured by firmware start/end timestamps. The gap is paired with the preceding acquisition, avoiding errors from subtracting unrelated time intervals.

Timing slopes and fixed engine costs

Fit T(delay) = intercept + slope x delay to the five measured settling settings. Slope checks how many settling waits occur per sweep. The zero-delay intercept is an extrapolated combined fixed cost; it does not isolate switch, transfer or USB time.

Engine	VMID	Measured slope us/us	Expected switches	Fixed intercept us	Max fit residual us
blocking	off	51.988	52	214.09	0.497
blocking	on	99.982	100	675.61	0.412
dma	off	51.985	52	398.01	0.550
dma	on	99.986	100	1306.18	0.487
lpspi	off	51.987	52	205.72	0.560
lpspi	on	99.991	100	639.80	0.263

Per address and per channel

For ADC order, each address visit yields one retained sensor value. Average address service is acquisition/50; period/50 is the amortized cost per sensor including shared work. Neither is the ADC conversion time. The ADC itself still uses a nominal 1 us serial frame; prime and VMID words are additional conversions.

Settle us	VMID	Amortized settle / channel us	Production acquisition / channel us	Production period / channel us	Diagnostic transfer scope / address us
1	off	1.04	5.131	5.151	3.330
1	on	2.00	14.777	14.800	13.189
2	off	2.08	6.173	6.200	3.318
2	on	4.00	16.780	16.800	14.165
5	off	5.20	9.300	9.320	3.326
5	on	10.00	22.767	22.790	17.166
10	off	10.40	14.479	14.501	3.334
10	on	20.00	32.767	32.790	22.166
50	off	52.00	56.081	56.103	3.318
50	on	100.00	112.766	112.789	62.165

Off: each address uses 2 ADC words; on: 7 words plus one VMID park/settle inside the transfer scope. Per-array sweep cost is approximately half the symmetric full-sweep work, with array handoff/parking costs assigned by the actual sequence. ADC-order array durations and individual GPIO edges were not separately timestamped.

Measured diagnostic phases

Existing teensy41_profile firmware; one repetition per setting, 1 s warm-up + 2 s measurement. Values below are mean phase times per whole sweep at 5 us settling, including warm-up. Profile scopes use the 600 MHz CPU cycle counter; interrupts remain enabled. Production firmware was restored afterward.

Engine	VMID	MUX prepare us	Session us	Transfer us	Validation us	Encode us	USB enqueue us	Profiler bookkeeping us
blocking	off	302.062	0.027	175.876	1.017	0.242	0.413	1.933
blocking	on	290.833	0.027	901.036	3.517	0.243	0.413	1.977
dma	off	303.080	0.027	358.087	1.167	0.243	0.420	1.966
dma	on	291.331	0.027	1535.470	4.084	0.243	0.422	2.105
lpspi	off	301.808	0.440	166.289	1.417	0.242	0.413	1.919
lpspi	on	290.562	0.441	858.288	4.334	0.243	0.415	2.019

Scopes overlap: validation is inside transfer; prepare/session/transfer are inside acquisition. Transfer with VMID on includes VMID park settling. Acquisition plus outside gap describes a period. Do not sum every phase column. Bookkeeping is diagnostic overhead, not a production component.

Engine	Off transfer / 16-bit word us	Validation / word us	Off prepare residual / switch us	On prepare residual / switch us
blocking	1.7588	0.0102	0.8089	0.8167
dma	3.5809	0.0117	0.8285	0.8266
lpspi	1.6629	0.0142	0.8040	0.8112

Per-word figures are aggregate diagnostic averages including loop/probe overhead, not oscilloscope CS-to-CS timings. Prepare residual subtracts the requested settling; it still includes 0.4 us explicit allowances, GPIO writes and scope overhead. VMID-on includes two no-op park calls in prepare, making its residual slightly different.

Settle us	VMID	Production period us	Profile period us	Probe/build slowdown
1	off	257.56	267.36	3.80%
1	on	740.02	756.92	2.28%
2	off	310.00	319.00	2.90%
2	on	840.00	856.00	1.90%
5	off	466.00	475.28	1.99%
5	on	1139.50	1156.20	1.47%
10	off	725.04	735.67	1.47%
10	on	1639.48	1655.82	1.00%
50	off	2805.15	2814.89	0.35%
50	on	5639.43	5655.91	0.29%

Probe overhead is measured as a whole-build difference. Subtracting adjacent cycle-read calibration from every phase would not remove all instrumentation work, so no such correction is claimed.

Per-word and per-address cost estimates

VMID-off transfer profiles provide two different operation counts: ADC order uses 100 words / 50 batches; address order uses 64 words / 14 batches. Fit transfer = words x word_cost + batches x batch_cost to all ten off profiles for each engine. This separates an estimated average word cost from per-address read-batch overhead under a uniform-cost assumption; these are diagnostic model estimates, not individual edge measurements.

Engine	Estimated word us	Read-batch overhead us	Prepare residual / switch us	Transfer model RMS us
blocking	1.7237	0.0736	0.8088	0.3375
dma	3.5342	0.0961	0.8250	0.2446
lpspi	1.6021	0.1209	0.8059	0.2858

Word cost includes the nominal 1 us ADC/serial frame, CS allowances, validation and driver/probe work. Switching residual is median((prepare minus requested settling)/actual changes) from VMID-off profiles across both orders; it includes the 0.4 us switch allowances, GPIO work and prepare probes. Neither component is corrected to an exact production-only electrical duration.

Order	Address group	VMID	Values	Words	Switch us	Settle us	Scan us	Service us	Per value us
ADC	each sensor-address visit	off	1	2	0.806	5.0	3.325	9.131	9.131
ADC	each sensor-address visit	on	1	7	1.612	10.0	11.456	23.068	23.068
Address	inputs 0-5	off	4	5	0.806	5.0	8.131	13.937	3.484
Address	inputs 0-5	on	4	10	1.612	10.0	16.262	27.874	6.969
Address	input 6 / MUX4 only	off	1	2	0.806	5.0	3.325	9.131	9.131
Address	input 6 / MUX4 only	on	1	7	1.612	10.0	11.456	23.068	23.068

LPSPI estimates at 5 us settling. Service includes the group's sensor switch, any VMID park, the requested waits and converter read batches. Off excludes the two additional shared mandatory park operations per sweep; both settings exclude sweep session, outer-loop work and USB. VMID wrapper-specific costs are approximated by the read-batch model, not independently isolated.

Each ADC has six four-value address groups and one single-value address group in address order; each has 25 single-value groups in ADC order. Divide a four-value service cost by four only to amortize work; this does not assign four simultaneous sample apertures. SPI conversions are sequential. Actual per-channel timestamps or exact array boundary intervals require extra instrumentation.

USB transmission and outside-acquisition costs

The timestamp gap after each acquisition covers frame encoding, the atomic USB enqueue, counters and main-loop/yield work. Production gaps are directly measured; encode and enqueue are separately measured only in the diagnostic build. Enqueue copies bytes to the USB transmit queue; it does not wait for host receipt.

Settle us	VMID	Production gap us	Profile encode us	Profile enqueue us	USB payload kB/s	Production gap / period
1	off	1.000	0.242	0.413	442.6	0.388%
1	on	1.169	0.243	0.413	154.1	0.158%
2	off	1.367	0.242	0.413	367.7	0.441%
2	on	1.000	0.243	0.413	135.7	0.119%
5	off	1.004	0.242	0.413	244.6	0.215%
5	on	1.121	0.243	0.415	100.0	0.098%
10	off	1.099	0.243	0.413	157.2	0.152%
10	on	1.118	0.243	0.415	69.5	0.068%
50	off	1.124	0.245	0.413	40.6	0.040%
50	on	1.129	0.245	0.413	20.2	0.020%

The firmware uses USB CDC, so the 460800 serial setting is not a UART bit-rate limit. Each frame is 114 bytes, with 100 bytes of sensor data (87.7% useful payload). The core submits full buffers immediately and flushes a partial transmit buffer with a nominal 75 us timer. That delay is asynchronous and is not an added 75 us wait in the sampling loop.

Pure serialization of 114 bytes would take 1.90 us at 480 Mbit/s or 76.0 us at 12 Mbit/s, before USB packet overhead and scheduling. These are illustrative lower bounds, not measured wire times or measured negotiated link speed. USB bus scheduling, driver buffering and host end-to-end latency were not instrumented and cannot be inferred from sample-rate slope.

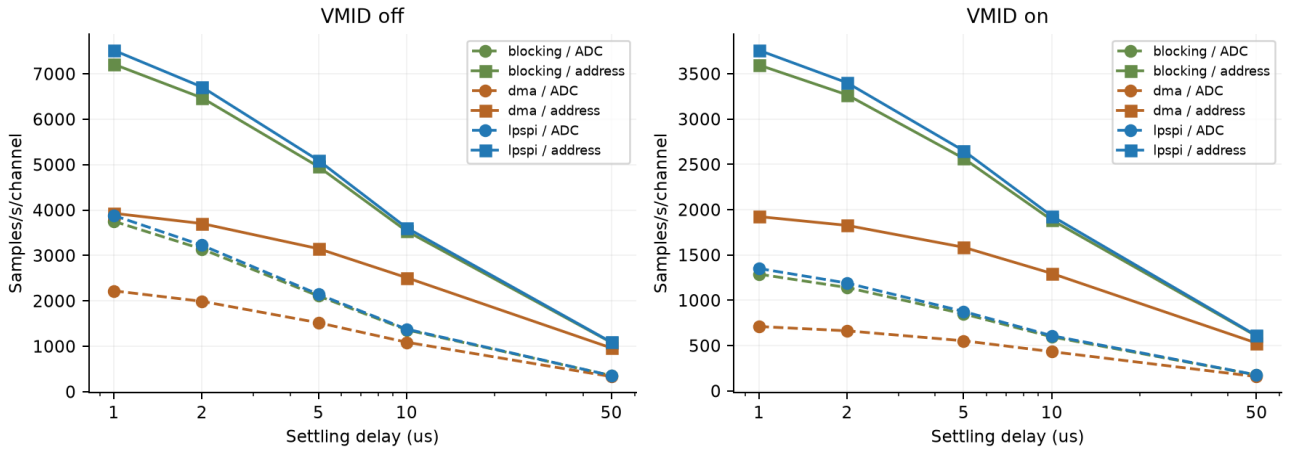
The useful optimization bound is the measured production gap: removing all encoding/enqueue/loop gap would save only that amount per sweep. USB interrupt servicing can also occur inside acquisition; its contribution is included in measured acquisition/remaining costs and is not individually resolved.

Sources: project UsbSerialController.cpp and usb_core_patch.py; pinned Teensy 1.162.0 core usb_serial.c (TRANSMIT_FLUSH_TIMEOUT, TX_SIZE and flush callback). No blocking binary flush is used.

Optimization control: group readings by address

Exploratory production control: scanorder mux, all three engines and the same five settling delays/VMID settings; one repetition per setting, 1 s warm-up + 3 s measurement. ADC-order primary results have three repetitions. This comparison establishes timing opportunity; different conversion history can change analog baseline or stimulated crosstalk.

Settle us	VMID	Blocking address Hz/ch	DMA address Hz/ch	LPSPI address Hz/ch	LPSPI gain vs ADC order
1	off	7,211.5	3,929.3	7,518.8	1.94x
1	on	3,596.8	1,924.1	3,759.4	2.78x
2	off	6,472.1	3,703.7	6,711.4	2.08x
2	on	3,268.0	1,827.0	3,401.4	2.86x
5	off	4,956.3	3,150.3	5,094.1	2.37x
5	on	2,566.4	1,585.7	2,651.3	3.02x
10	off	3,537.7	2,512.6	3,610.1	2.62x
10	on	1,886.8	1,297.0	1,930.5	3.17x
50	off	1,084.8	963.9	1,091.4	3.06x
50	on	606.2	528.9	610.8	3.44x



Off: address grouping removes 36 settling events and 36 selector-prime words per sweep. On: it removes 72 settling events and 216 total words. No higher SCLK or concurrent ADC acquisition is required. Scan order changes are already supported; no firmware modification was made.

Baseline context for address grouping

The two orders are not electrically equivalent histories. Table uses median channel baselines in the post-warm-up record; primary ADC-order values are medians across three repetitions, address-order control values come from one repetition. These are whole-record comparisons, not a new burst-masked noise study.

Settle us	VMID	Array A spread counts ADC / address	Array B spread counts ADC / address	Worst channel SD counts ADC / address
1	off	41.0 / 66.0	207.0 / 182.0	0.495 / 0.899
1	on	35.0 / 33.0	55.0 / 93.0	0.496 / 0.500
2	off	42.0 / 67.0	181.0 / 159.0	0.497 / 0.500
2	on	35.0 / 33.0	59.0 / 89.0	0.499 / 0.546
5	off	44.0 / 68.0	133.0 / 121.0	0.500 / 0.499
5	on	36.0 / 34.0	63.0 / 78.0	0.500 / 0.500
10	off	41.0 / 66.0	89.0 / 85.0	0.493 / 0.500
10	on	36.0 / 34.0	60.0 / 65.0	0.498 / 0.500
50	off	16.0 / 38.0	11.0 / 12.0	0.497 / 0.500
50	on	29.0 / 27.0	12.0 / 12.0	0.500 / 0.444

Baseline spread is the range across 25 channel median levels, not temporal noise. Counts x 3.3/4096 converts these measurements to volts. Individual control-channel metrics can be recomputed from the hashed raw frames and exact payload_routes in each stopped status snapshot.

A faster setting is a useful candidate only if acceptable analog settling, ghosting and response are maintained with sensors attached. Open-connector baseline measurements alone cannot establish those properties. The original quiet-noise report remains unchanged; this supplement targets timing.

Optimization priorities and likely savings

5 us settling / VMID off

Measured change	Time saved per sweep	Rate change
DMA to LPSPI	192.00 us	41.2%
Blocking to LPSPI	8.00 us	1.7%
LPSPI ADC to address order	269.70 us	2.37x

5 us settling / VMID on

Measured change	Time saved per sweep	Rate change
DMA to LPSPI	666.70 us	58.5%
Blocking to LPSPI	35.62 us	3.1%
LPSPI ADC to address order	762.32 us	3.02x

1. Keep LPSPI for current ADC-order timing. It retains one SPI transaction per sweep and directly transfers 16-bit words. Blocking enters/exits an SPIClass transaction and changes frame format per word. DMA configures/completes a two-byte asynchronous transfer per word, including event handling and yield. With the present implementation, DMA is not a speed improvement.

2. Prefer the existing address-group order if analog behavior permits. This has the largest measured opportunity without changing firmware. It amortizes one external switch/settle and one selector prime over up to four retained values. Validate stimulated ghosting and channel response before changing the operational scan order.

3. Investigate burst transfers within a fixed address group. A driver could keep CS low for multiple 16-clock conversions and pipeline a prepared command sequence, reducing per-word setup, CS delays and DMA launches. Firmware currently toggles CS for every word. At ADC order, consolidating each 2-word sensor batch and each 5-word VMID batch would reduce explicit CS allowances by 10 us/sweep off or 50 us/sweep on, before any software saving. These are calculations, not measured optimized-driver results.

4. Consider selector primes conditionally, with validation. In VMID-off ADC order, consecutive external inputs within one MUX keep the same ADC selector. If electrical settling and next-conversion state allow omitting redundant primes, eight MUX starts per sweep would need primes rather than all 50, removing up to 42 conversion words. That changes conversion history and has not been implemented or electrically validated.

5. Reduce GPIO/delay-call overhead only after the larger changes. Cache GPIO masks or use grouped address writes while preserving disable, turn-off/turn-on allowances and EN polarity. Do not reduce analog settling merely because switching CPU overhead is small. USB encoding/enqueue is a small measured share at these rates; prioritize it only if counters show transport pressure.

Multiple back-to-back 16-clock conversions with CS low are supported by the TI serial interface. Burst-driver feasibility follows that specification; safe timing, payload identity, bounds/timeouts, cleanup and analog history still require tests. TI ADC124S101, Using the ADC / Power Management.

Integrity, evidence and limits

180 captures completed: 90 primary ADC-order production, 30 address-order production controls, 60 diagnostic profiles. 1,472,776 full sweeps received (73,638,800 transmitted channel values, including warm-up). 1,472,778 acquired sweeps; 2 explicitly accounted whole-sweep USB discards. Received equals sent for every capture; acquired equals sent plus discarded.

All raw streams were independently decoded with the existing incremental frame parser and structured frame reader; samples/timing/counts agreed. SHA-256 hashes were rechecked during report analysis. ADC values stayed within 0-4095, timestamps were monotonic after wrap handling, and every acquisition ended before the next one started. Firmware error counters remained zero.

Production tests used ordinary teensy41 firmware. The existing teensy41_profile build was temporarily loaded for phase measurement and then ordinary production firmware restored. No acquisition source or wire format was changed. The recorder saves stopped status snapshots, profiling summaries and command logs. Final board status is saved in firmware_lifecycle.json; original full-array, blocking, MUX-order, 16 MHz, repeat-1, 5 us, VMID-off settings were restored and the board left stopped.

What is measured, calculated or still unresolved

Quantity	Evidence / interpretation
Sweep period, useful sampling rate	Production device start timestamps; after warm-up
Total acquisition and outside gap	Production start/end timestamps; correctly paired
Address/word/event counts	Firmware plan plus host replay tests
Requested settling and explicit allowances	Event counts x configured firmware delays
ADC track / conversion / shift	Clock-derived datasheet components; overlap
Prepare / transfer / session / encode / enqueue	Cycle-counter profiles; probes included
Electrical MUX propagation / analog settling curve	Not individually measured by this experiment
Per-channel aperture / exact sample timestamps	Not transmitted; only shared sweep start/end interval
USB physical transfer / host end-to-end latency	Not independently measured; enqueue overlaps acquisition

Rate slope alone cannot identify every independent time component: it reveals the number of settling waits and a combined intercept. Diagnostic scopes narrow that intercept, but exact electrical switch latency, exact SCLK waveform and host transport latency would require synchronized external measurements. No unsupported precision is assigned to those quantities.

CSV evidence: production_capture_metrics.csv (120 rows), production_summary.csv (60 settings), diagnostic settings: 60, delay_fits.csv (36 fits) and timing_analysis.json. The original settling report and its 30 captures provide an independent prior rate/baseline reference.